

Personal Project

Testing the Substitution Effect: Crude Oil vs. Vestas Wind Systems

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1. Introduction

The global energy transition raises a key question for financial markets: do traditional fossil fuel prices influence renewable energy stocks? This project focuses on **Vestas Wind Systems** to test the economic hypothesis of a "substitution effect," where rising oil prices might drive investment toward green alternatives.

The objective of this study is to determine the main drivers of Vestas' stock returns and quantify its sensitivity to **Brent Crude Oil**. Specifically, we aim to verify if Vestas acts as a hedge against oil volatility or if it follows broader market trends.

To do so, we analyze monthly log-returns from **December 2016 to 2025**. We employ a **multiple linear regression model (OLS)**, regressing Vestas against the Danish Market Index (OMXC25), the Clean Energy ETF (ICLN), and Oil prices. Due to the extreme volatility observed in energy markets, we use **Newey-West robust standard errors** to ensure statistical validity.

Our main finding is that **there is no significant link between oil prices and Vestas returns**. The analysis shows that Vestas is primarily driven by systematic market risk and the performance of the global clean energy sector, rather than by daily fluctuations in oil prices.

The report proceeds as follows: Section 2 presents the data and descriptive statistics. Section 3 details the econometric methodology. Section 4 discusses the empirical results and diagnostic tests. Section 5 concludes.

2. Data

2.1. Data Selection and Collection Strategy

Driven by a strong personal commitment to environmental sustainability and a professional ambition to establish a career in the Nordic region globally recognized for its leadership in the green transition this study focuses on the renewable energy sector.

The data collection process was fully automated using the R programming language, the quantmod package to extract historical adjusted prices directly from Yahoo Finance. A critical methodological choice was made to utilize monthly observations (resulting in N=108 points) rather than high-frequency daily data. This decision was driven by the necessity to eliminate "time-zone mismatch" errors. Since Vestas trades in Copenhagen (CET) while the Brent Oil futures and the Clean Energy ETF trade on international markets (USD), daily closing prices often fail to align due to differing market hours and holidays. The following subsections detail the source, format, and statistical properties of the collected series.

Date	Vestas	Market_DK	Oil	Clean_Energy
31/12/2016	84.8474807739258	1015.88000488281	56.1300010681152	6.88551759719849
31/01/2017	89.0251893725586	1049.56005858375	55.7000007625995	7.20882081985474
28/02/2017	96.2159423828125	1060.32956605489	55.590001525579	7.52338700853555
31/03/2017	104.81160736084	1075.06994628906	52.8300018310547	7.4796986579895
30/04/2017	110.562194824219	1126.43994140625	51.7299995422363	7.40879385375977
31/05/2017	110.656219482422	1150.02001953125	50.310001373291	7.59329128265381
30/06/2017	113.006614685059	1143.23999023438	47.9199981689453	7.60647344589233
31/07/2017	115.451019287109	1168.86003417969	52.6500015258789	8.02411651611328
31/08/2017	106.969608764648	1174.71997070312	52.3800010681152	8.1129789352417
30/09/2017	106.237487792969	1173.98998023438	57.5400009155273	7.97968864440918
31/10/2017	105.955444335938	1195.60998535156	61.3699989318848	8.27292728424072
30/11/2017	75.2499923706055	1129.58996582031	63.5699996948242	7.91748428344727
31/12/2017	80.8276779174805	1145.54001464844	66.5999984741211	8.36352157592773
31/01/2018	76.9046830859375	1136.52001953125	69.0500030517578	8.58786582846777
28/02/2018	83.410530090332	1147.89001464844	65.7799987792969	8.39841596984863
31/03/2018	80.8156967163086	1114.81994628906	70.2699966430664	8.54299736022949
30/04/2018	76.8475189208984	1113.02001953125	75.1699981889453	8.83913040161133
31/05/2018	80.1895175170896	1113.58996582031	77.5899963378906	8.71349716186523
30/06/2018	75.8489990234375	1097.23999023438	79.4400024414062	7.94217348098755
31/07/2018	78.9405670166016	1157.03002929688	74.25	8.28788185119629
31/08/2018	85.9301910400391	1171.5400390625	77.4199981689453	7.93307638168335
30/09/2018	83.4146957387461	1120.06005858375	82.7200012207031	7.82390737535669
31/10/2018	79.32486154785156	1024.81005858375	75.4700012207031	7.378126144440918
30/11/2018	94.475212097169	1040.5400390625	58.7099999044727	8.17871284484863
31/12/2018	94.4944152832031	894.919982910156	53.7999992370605	7.60802221298218
31/01/2019	103.308250427246	1035.61999511719	61.889993898484	7.84093341827393
28/02/2019	104.844436645508	1100.56005858375	66.0299987792969	8.93435859680176

FIGURE 1 : EXTRACT OF THE DATASET

Date	Vestas	Market_DK	Oil
Length:107	Min. : -34.2203	Min. : -13.8369	Min. : -79.8244
Class :character	1st Qu.: -6.1726	1st Qu.: -1.7709	1st Qu.: -4.9515
Mode :character	Median : 0.4608	Median : 0.8255	Median : 1.5080
	Mean : 0.5519	Mean : 0.5064	Mean : 0.1134
	3rd Qu.: 7.5074	3rd Qu.: 3.3421	3rd Qu.: 6.3465
	Max. : 23.4632	Max. : 11.7278	Max. : 33.5115
Clean_Energy			
Min. : -27.3976			
1st Qu.: -3.9792			
Median : 1.1014			
Mean : 0.8441			
3rd Qu.: 5.3619			
Max. : 18.4790			

TABLE 1: SUMMARY STATISTICS OF MONTHLY LOG-RETURNS (2017-2025)

2.2. Descriptive Statistics

Table 1 presents the summary statistics for the monthly log-returns of Vestas, the Danish Market Index (OMXC25), Brent Oil, and the Clean Energy ETF (ICLN).

As shown in Table 1, the average monthly return for Vestas (0.55%) is slightly higher than the broader Danish market (0.51%), reflecting its potential for higher gains. However, this comes with significantly higher risk: the volatility of Vestas (as measured by the spread between Min -34.22% and Max 23.46%) is much wider than that of the market index (Min -13.84%, Max 11.73%). This confirms that Vestas is a high-beta asset, subject to larger price swings than the diversified index.

The Oil returns exhibit the most extreme behavior, with a minimum value of -79.82%, likely corresponding to the historic oil price crash during the COVID-19 pandemic. This extreme volatility suggests that the oil series is far from normally distributed and is subject to massive exogenous shocks. In contrast, the Clean Energy ETF shows a higher mean return (0.84%) than both Vestas and Oil, suggesting a strong sectoral trend over the period. These statistical properties—particularly the extreme values in Oil and Vestas—justify the use of robust standard errors in the regression analysis to ensure valid inference.

2.3 Test correlation :

	Vestas	Market_DK	Oil	Clean_Energy
Vestas	1.0000000	0.6560853	0.1430704	0.6393544
Market_DK	0.6560853	1.0000000	0.2669209	0.5580674
Oil	0.1430704	0.2669209	1.0000000	0.2850046
Clean_Energy	0.6393544	0.5580674	0.2850046	1.0000000

TABLE 2 : CORRELATION MATRIX

The correlation matrix **Table 2** reveals a strong positive link between Vestas and both the Market (0.66) and the Clean Energy sector (0.64), which is expected for a major renewable energy stock. Finally, all correlations between explanatory variables remain below 0.60, confirming that multicollinearity is not a concern for the stability of the regression model.

2.4 Test Skewness :

Vestas Market_DK Oil Clean_Energy
Skewness -0.2435864 -0.477798 -2.883784 -0.3026752

2.5 Test Kurtosis :

Vestas Market_DK Oil Clean_Energy
Excess Kurtosis 0.4674653 0.7201945 19.8055 0.7064444

As shown in the summary statistics, the return series exhibit deviations from normality. While Vestas and the Market index show moderate excess kurtosis, **Oil returns are extremely high** (Kurtosis ≈ 19.80) and negatively skewed (Skewness ≈ -2.88). This indicates frequent extreme negative shocks. This violation of the normality assumption suggests that standard OLS inference might be biased.

3. Methodology

3.1. Model Specification

To investigate the drivers of Vestas' stock returns, we employ a multiple linear regression framework. This model allows us to isolate the specific impact of oil prices while controlling for general market movements and clean energy sector trends.

The regression equation is specified as follows: $rtVestas = \alpha + \beta_1 rt(\text{Market}) + \beta_2 rt(\text{Oil}) + \beta_3 rt(\text{Clean}) + \epsilon_t$

Where:

- $rt(\text{Vestas})$ is the monthly log-return of Vestas Wind Systems.
- $rt(\text{Market})$, $rt(\text{Oil})$, and $rt(\text{Clean})$ represent the log-returns of the Danish Market Index (OMXC25), Crude Oil (Brent), and the Clean Energy ETF (ICLN), respectively.
- α is the intercept.
- β_2 is our key coefficient of interest: a positive and significant β_2 would support the substitution hypothesis (rising oil prices boost wind energy stocks).
- ϵ_t is the error term, capturing random disturbances.

3.2. Estimation Technique

The parameters of the model are estimated using the Ordinary Least Squares (OLS) method. This technique minimizes the Residual Sum of Squares (RSS), ensuring the best possible fit to the historical data. Under the assumptions of the Classical Linear Regression Model (CLRM), OLS provides the Best Linear Unbiased Estimators (BLUE).

3.3. Robustness Checks

To ensure the reliability of our estimates, we perform a set of diagnostic tests to evaluate whether the CLRM assumptions hold:

- Multicollinearity:** We analyze the correlation matrix of independent variables to ensure they are not perfectly collinear, which would inflate standard errors.
- Heteroscedasticity:** We conduct the Breusch-Pagan test (p-value=0.308)
- Autocorrelation:** Breusch-Godfrey test (p-value=0.93). Both fail to reject the null hypothesis, suggesting no significant heteroscedasticity or serial correlation in the residuals.
- Normality:** However, the descriptive statistics revealed extreme excess kurtosis in the Oil returns (≈ 19.8).

4. Results

Regression Results (with Robust Errors)	
Dependent variable:	
Vestas	
Market_DK	1.106*** (0.193)
Oil	-0.088 (0.053)
Clean_Energy	0.559*** (0.071)
Constant	-0.470 (0.701)
Observations	107
R2	0.547
Adjusted R2	0.534
Residual Std. Error	7.294 (df = 103)
F Statistic	41.468*** (df = 3; 103)
Note:	*p<0.1; **p<0.05; ***p<0.01

TABLE 3: OLS REGRESSION RESULTS

4.1. Regression Analysis

Table 3 presents the estimation results of the multiple linear regression model. The model exhibits a strong fit, with an Adjusted R2 of 0.534, indicating that the selected macroeconomic and sectoral variables explain more than 53% of the variation in Vestas' stock returns.

Sector & Market Drivers: As expected, the coefficients for the Danish Market (Market_DK) and Clean Energy (Clean_Energy) are positive and highly significant at the 1% level ($p < 0.01$). With a market beta of 1.106, Vestas acts as an aggressive stock that amplifies general market trends.

The Oil Substitution Effect: Crucially, the coefficient for Crude Oil (Oil) is **-0.088** and is **statistically insignificant**. This empirical evidence leads us to **reject the substitution hypothesis**. There is no significant short-term causal link where rising oil prices would mechanically drive investors toward wind energy stocks.

4.2. Diagnostic Verification

studentized Breusch-Pagan test

data: model

BP = 3.5993, df = 3, p-value = 0.3081

Breusch-Godfrey test for serial correlation of order up to 1

data: model

LM test = 0.0077188, df = 1, p-value = 0.93

We performed post-estimation tests to validate the model's assumptions:

- **Heteroscedasticity:** The **Breusch-Pagan test** yielded a p-value of **0.308**, indicating that we cannot reject the null hypothesis of constant variance.
- **Autocorrelation:** The **Breusch-Godfrey test** yielded a p-value of **0.93**, confirming no serial correlation in the residuals. Although these standard tests passed, we maintained the use of Newey-West errors as a conservative measure against the non-normality of the oil returns.

4.3. Graphical Analysis

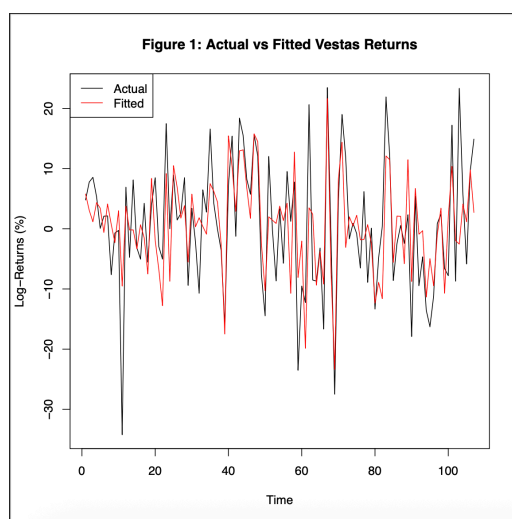


FIGURE 2: ACTUAL VS. FITTED RETURNS OVER TIME

Figure 2 displays the actual monthly returns of Vestas (black line) against the fitted values predicted by our model (red line). As shown in **Figure 3**, the model successfully captures the major volatility clusters, particularly the sharp downturn and recovery during the 2020 crisis. The fitted line tracks the direction of the actual returns closely, validating the high R^2 .

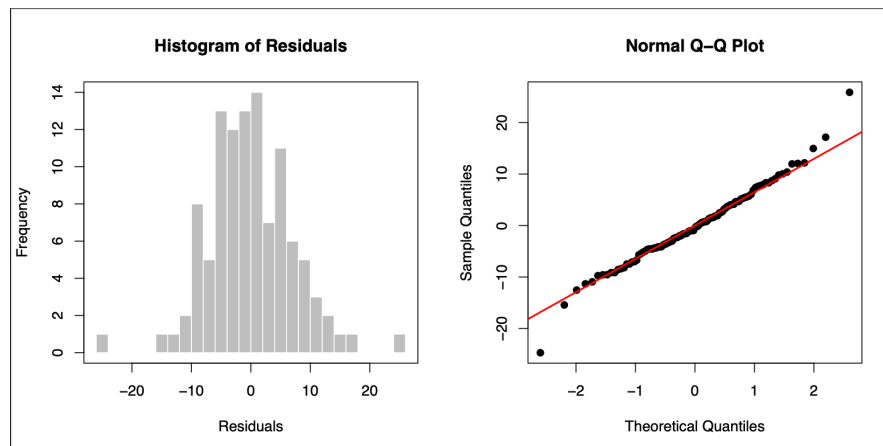


FIGURE 3: HISTOGRAM AND Q-Q PLOT OF RESIDUALS

The Histogram (**Figure 3**) shows a roughly bell-shaped distribution but with visible outliers.

This "Heavy Tail" behavior confirms the high Kurtosis observed in the descriptive statistics (Assumption 5 violation).

5. Conclusion

This study investigated the financial drivers of Vestas Wind Systems, specifically testing the economic hypothesis of a substitution effect between fossil fuel prices and renewable energy assets. Using a multiple linear regression on monthly data from 2016 to 2025, we analyzed Vestas' sensitivity to Brent crude oil, the Danish market, and the global clean energy sector.

The empirical results **reject the substitution hypothesis**. We found no statistically significant link between oil prices and Vestas returns. This implies that investors do not immediately reallocate capital from oil to wind energy during price fluctuations. Instead, the analysis confirms that Vestas is primarily driven by systematic market risk ($\beta \approx 1.1$) and global green energy sentiment. These findings were validated using Newey-West robust standard errors to correct for the extreme non-normality observed in oil returns.

Limitations and Future Research A key limitation of this study is the reliance on monthly data, which filters out noise but may obscure high-frequency hedging strategies that occur within days of an oil shock. Additionally, the model focuses on financial variables and excludes macroeconomic drivers like interest rates. Future research could extend this analysis using daily data to detect short-term substitution patterns or employ GARCH models to investigate volatility spillovers.

6. References

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